TECHNICAL DESIGN DOCUMENT

🏛️ Sahayak AI (सहायक AI)

Government Welfare Scheme Advisor & Explainable Match Engine

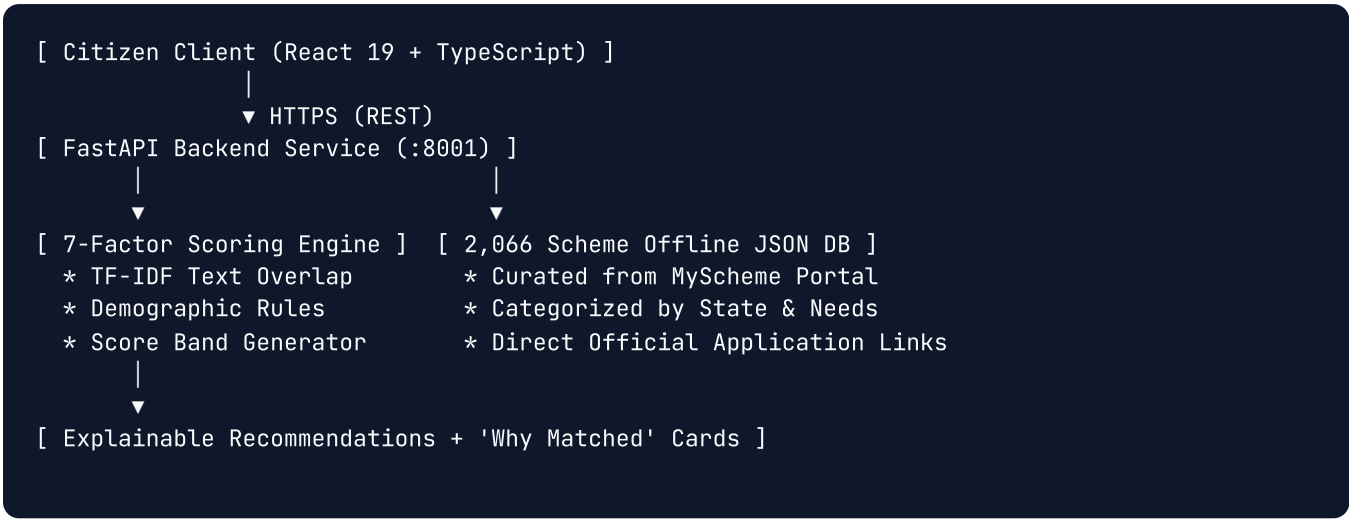
Team: Analog Roses | **Event:** Hack for Social Cause | **Date:** October 2026

Executive Summary: Sahayak AI is a privacy-first digital public service discovery tool that eliminates welfare exclusion by indexing **2,066 Indian Central and State government schemes** over an explainable 7-dimension scoring engine. It collects *zero personally identifiable information (no Aadhaar, no bank accounts)* and transparently reveals why each scheme was recommended before directing the citizen to official portals.

1. Problem Space & Need Analysis

- **Information Fragmentation:** Welfare policies are spread across dozens of departmental silos with varying guidelines.
- **Exclusion Errors:** Millions of eligible citizens never apply because eligibility rules are perceived as opaque.
- **Data Privacy Hesitation:** Asking citizens for Aadhaar or bank passbooks creates apprehension and fraud risk.
- **Hallucination Hazards:** Unconstrained generative AI models frequently invent false grants or incorrect eligibility criteria.

2. System Architecture

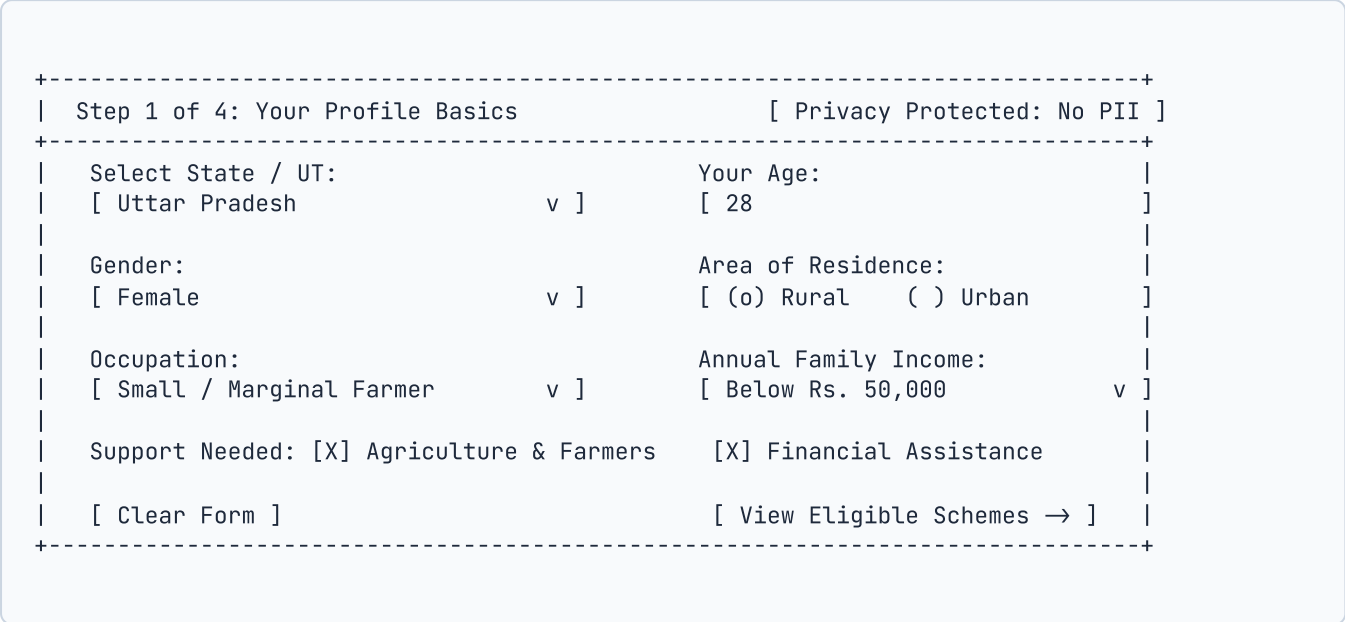


3. The 7-Factor Scoring Matrix

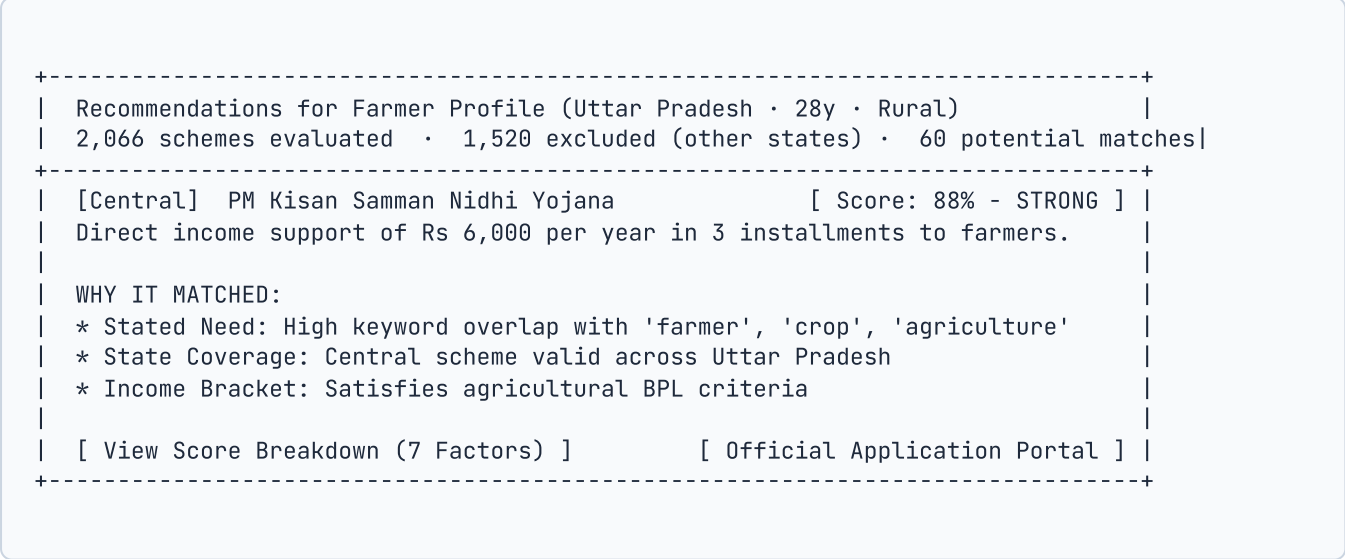
Dimension	Weight	Description & Method
Needs Relevance	**35%**	TF-IDF vector text similarity between stated citizen support needs and scheme descriptions/tags.
State Coverage Fit	**20%**	100% for state matches; 85% for national Central schemes; excludes schemes of other states.
Age Boundary Fit	**15%**	Validates numerical age brackets and seniority criteria against citizen age.
Income Fit	**10%**	Matches BPL / low-income brackets with government subsidy rules.
Gender Focus	**10%**	Prioritizes women and girl-child specific programs when beneficiary is female.
Social Category	**5%**	Detects SC, ST, OBC, or minority priority criteria.
Residence Fit	**5%**	Tailors recommendations to Rural vs. Urban targeted programs.

4. User Interface Wireframes

4.1 Citizen Intake Wizard (/find)



4.2 Recommendation Results & Explainability (/results)



5. Privacy, Security & Disclaimers

- **Zero PII Intake:** Never requires Aadhaar, phone numbers, bank accounts, or documents.
- **Truth Anchoring:** All recommendations are preliminary; users are directed to official portal links (e.g. `myscheme.gov.in`, `pmkisan.gov.in`).

- **Deterministic Matcher:** Avoids generative AI hallucinations by relying on explainable TF-IDF text metrics and objective rules.